Stirling Engine Study Summary

Given Parameters (from code/report)

Parameter	Symbol	Value	Units
Power piston crank	r_p	0.025	m
Power connecting rod	ℓ_p	0.075	m
Displacer crank	r_d	0.020	m
Displacer rod	ℓ_d	0.140	m
Displacer volume	V_d	4.0×10^{-5}	m^3
Cylinder bore	D	0.050	m
Compression ratio	CR	1.70	—
Temps (hot/cold)	T_h/T_c	900/300	K
BDC pressure (abs)	P_{BDC}	500	kPa
Atmospheric (abs)	P_{atm}	101.3	kPa
Average speed	$\overline{\Omega}$	650	rpm
Coeff. fluctuation	C_f	0.003	—
Flywheel (w, t, ρ)		0.025, 0.050, 7750	m, m, kg/m^3

Core Kinematics

Cross-sectional area: $A = \pi D^2/4$. Rod-obliquity and slider-crank for position at angle θ and crank r, rod ℓ:

$$\beta(\theta) = \arcsin\left(\frac{r}{\ell}\sin\theta\right), \qquad x(\theta) = \ell\cos\beta(\theta) - r\cos\theta. \tag{1}$$

Displacer is phase-shifted by ϕ: $x_d(\theta) = x(\theta + \phi)$, power piston $x_p(\theta) = x(\theta)$.

Volumes

Cold and hot heights and volumes:

$$h_c(\theta) = [x_d(\theta+\phi) - x_p(\theta)] - h_{\text{pin}} - \tfrac{1}{2}h_d, \qquad V_c(\theta) = A\,h_c(\theta), \tag{2}$$

$$h_h(\theta) = H_{\text{tot}} - \tfrac{1}{2}h_d - x_d(\theta+\phi), \qquad V_h(\theta) = A\,h_h(\theta). \tag{3}$$

Regenerator volume V_r is constant.

Schmidt Analysis (isothermal, ideal gas)

Total gas mass at BDC:

$$m \;=\; \frac{P_{\text{BDC}}}{R}\left(\frac{V_c(0)}{T_c} + \frac{V_r}{T_r} + \frac{V_h(0)}{T_h}\right).$$

Instantaneous pressure:

$$P(\theta) \;=\; \frac{mR}{\dfrac{V_c(\theta)}{T_c} + \dfrac{V_r}{T_r} + \dfrac{V_h(\theta)}{T_h}}.$$

Net axial force on power piston: $F_p(\theta) = (P(\theta) - P_{\text{atm}})A$. Torque:

$$\tau(\theta) \;=\; -\,F_p(\theta)\,\frac{r_p\,\sin\theta}{\cos\beta(\theta)}.$$

Cycle Work, Power, Efficiency

Work per cycle (trapz over angle): $W = \oint \tau \, d\theta \approx \sum_k \tau(\theta_k) \, \Delta\theta$. Power: $P = \overline{\tau} \, \omega_{\text{avg}}$. Ideal Stirling work reference: $W_{\text{ideal}} = mR \, (T_h - T_c) \, \ln(V_{\max}/V_{\min})$. Efficiency: $\eta = W/W_{\text{ideal}}$.

Flywheel Sizing

Torque deviation $T_{\text{dev}} = \tau - \overline{\tau}$, energy fluctuation ΔE = peak-to-peak of $\int T_{\text{dev}} \, d\theta$. Required inertia:

$$I_{\text{req}} = \frac{\Delta E}{C_f \, \omega_{\text{avg}}^2}.$$

Thick-rim model with outer radius R and inner $R_{\text{in}} = R - t$:

$$I_{\text{rim}} = \tfrac{1}{2} M(R) \, (R^2 + R_{\text{in}}^2), \quad M = \rho \, \pi w \, (R^2 - R_{\text{in}}^2).$$

Solve $I_{\text{rim}} = I_{\text{req}}$ via fixed-point update $R \leftarrow R \, \eta^{1/3}$ with $\eta = I_{\text{req}}/I_{\text{act}}$.

Dynamics

Steady load torque $T_{\text{load}} = \overline{\tau}$. Using work-energy with cumulative work $W_{\text{net}}(\theta) = \int (\tau - T_{\text{load}}) \, d\theta$:

$$\Omega^2(\theta) = \Omega_{\text{avg}}^2 + \frac{2W_{\text{net}}(\theta)}{I}, \quad \Omega = \sqrt{\cdot}\,; \text{ then normalize to maintain } \overline{\Omega}.$$

Phase Optimization (three-stage, to 0.01°)

Coarse: $\phi \in [30°, 150°]$ step 2°; Medium: ±6° around best at 0.1°; Fine: ±0.5° at 0.01°. Select ϕ maximizing $\overline{\tau} \, \omega_{\text{avg}}$.

Key Results (from run)

- Volume ranges (total/hot/cold): 140.25–238.42 cm^3; 32.26–110.80 cm^3; 35.61–163.18 cm^3; CR=1.70.
- Pressure range/mean/ratio: 474.63–1178.83 kPa, 730.39 kPa, ratio 2.48.
- Torque: min/max −24.841/39.807 N m; mean 3.648 N m; work/cycle 22.982 J.
- Ideal work 94.905 J; efficiency 24.2%.
- Power (mean-torque method) 248.28 W at 650 rpm.
- Flywheel: $I_{\text{req}} = 4.4787 \text{kg m}^2$; $D_{\text{out}} = 0.887$m; $D_{\text{in}} = 0.787$m; mass 25.47 kg.
- Dynamics: RPM range 648.9–650.9; C_f actual 0.0030; $T_{\text{load}} = 3.648$N m.
- Phase optimization: optimal $\phi = 103.6°$; max power 255.80 W; mean torque at optimum 3.758 N m.

Figures to Review

P–v, torque vs. angle, pressure vs. angle, volumes vs. angle, angular velocity vs. angle, energy vs. phase. See PNGs in project root.

Common Pitfalls / Checks

- Use absolute pressure in Schmidt; subtract P_{atm} only for net force.
- Ensure consistent units (m, Pa, N, J, rad/s). Convert rpm to rad/s for ω.
- Phase optimization requires coarse-to-fine narrowing to reach 0.01°.
- Normalize speed profile after energy-based integration to match target average.
- Flywheel iteration: check convergence tolerance and physical diameter limits.